Apertus Difference Ranker - Short Method Description

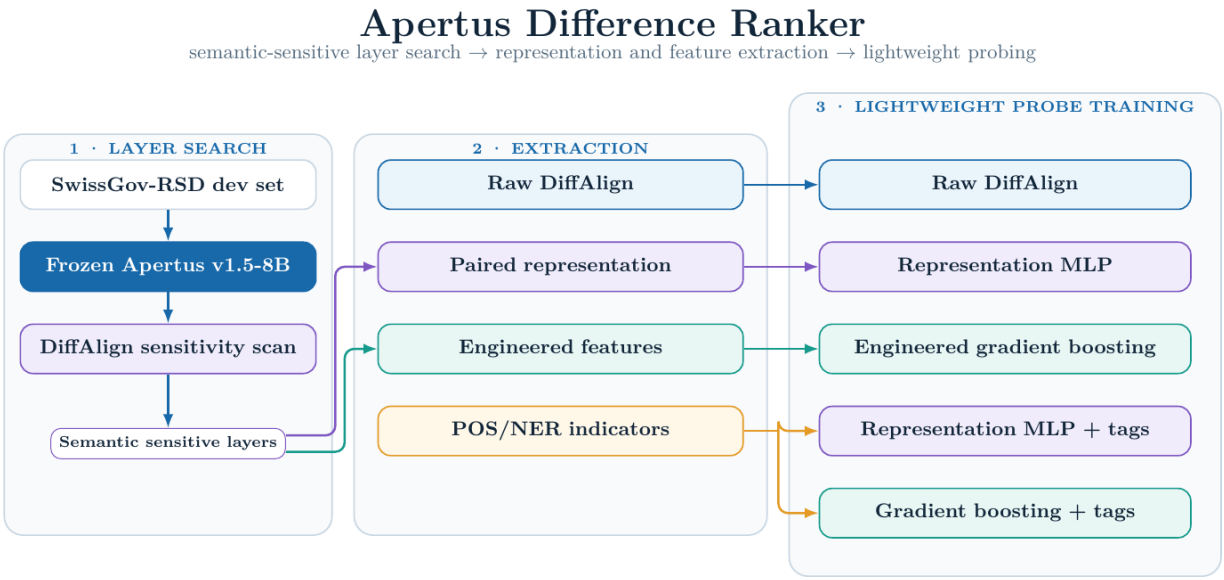
Track: Hack Apertus Track 2A - UZH SwissGov-RSD

Task: Rank token-level semantic differences in related multilingual government documents

Primary backbone: swiss-ai/Apertus-v1.5-8B

Overview

Apertus Difference Ranker combines frozen Apertus representations with small supervised probes. The method first searches for semantic-sensitive Transformer layers, then extracts token-level cross-document evidence, and finally fits lightweight rankers on the SwissGov-RSD development split. Apertus is never fine-tuned, and the official test split is used only for the final evaluation.



1. Semantic-sensitive layer search

Each side of a bilingual document pair is encoded independently with frozen Apertus. SwissGov whitespace tokens are mapped to Apertus subwords and pooled back to the benchmark tokenization. At every Transformer layer, raw DiffAlign assigns source token i the score

```
d_l(i) = 1 - max_j cosine(h_l(a_i), h_l(b_j)).
```

The operation is repeated in both language directions. Layer sensitivity is measured with macro Spearman on the 453-document fitting partition only. The 8B system selects layer 14 for paired representations and layers 14-18 for engineered features. The controlled 70B run selects representation layer 17 and engineered layers 17, 19, 18, 16, and 20.

2. Representation and feature extraction

The project evaluates two trainable branches:

1. **Paired-representation MLP.** Each source state is paired with its cosine-nearest target state. The probe receives the source state, target state, absolute difference, and elementwise product. A two-layer MLP maps this vector to a continuous difference score in [0, 1].
2. **Engineered gradient boosting.** For each of five selected layers, the system computes eight alignment statistics: nearest cosine distance, mean top-5 distance, top-1/top-2 margin, normalized alignment entropy, cosine-row mean, cosine-row standard deviation, mutual-nearest-neighbour status, and source-state norm. Local-context distances and four static token features produce a 54-dimensional vector for histogram gradient boosting.

Both branches also have tagged variants that append 23 fixed-width indicators: six coarse NER categories and seventeen Universal POS categories. These tags come from frozen auxiliary taggers; all semantic representations and alignment features come from Apertus.

3. Lightweight fitting and controlled evaluation

The 504 SwissGov-RSD development documents are split by document and language into 453 fitting documents and 51 validation documents. Positive-difference tokens receive higher sample weight to reduce label imbalance. The MLP uses a weighted regression and ranking objective with validation-based early stopping. The gradient booster uses 300 iterations with fixed hyperparameters. Layer selection, fitting, early stopping, and model choice use development data only.

The frozen models are then applied once to all 168 official test documents, covering 132,080 non-punctuation tokens. Performance is the unweighted mean of token-level Spearman correlations for English-German, English-French, and English-Italian.

Held-out results

Backbone and method	Test macro Spearman	Test pooled Spearman
8B raw DiffAlign	0.2632	0.2672
8B paired-representation MLP	0.3092	0.3199
8B engineered gradient boosting	0.3247	0.3296
8B paired-representation MLP + POS/NER	0.3190	0.3250
8B engineered gradient boosting + POS/NER	**0.3275**	**0.3325**
70B raw DiffAlign	0.2737	0.2786
70B best lightweight probe	0.3168	0.3244

The submitted 8B system improves macro Spearman by **0.0643 absolute** over raw 8B DiffAlign. The 70B backbone slightly improves raw alignment, but its best probe does not surpass the smaller model. The final system therefore favors the 8B engineered-feature ranker with POS/NER as the strongest and most efficient configuration.

Reproduction

Run the frozen result replay from the track_2a directory:

```
make run
```

Full GPU reproduction instructions, exact model revisions, and Clariden job commands are provided in docs/REPRODUCE.md. A detailed discussion of architecture, evaluation, and limitations is available in technical_report.md and technical_report.pdf.